

## Unit 2 Classwork — Exponent Rules

Exponent Rules · Power/Zero/Negative Exponents

### A. Exponent Rules — Product & Quotient

#### Problem 1

Simplify:  $x^4 \cdot x^3$

$$\underline{x^7}$$

#### Problem 2

Simplify:  $y^8 \div y^3$

$$\underline{y^5}$$

#### Problem 3

Simplify:  $(2x^3)(3x^2)$

$$\underline{6x^5}$$

#### Problem 4

Simplify:  $15a^6 \div 3a^2$

$$\underline{5a^4}$$

#### Problem 5

Simplify:  $a^2 \cdot a^5 \cdot a$

$$\underline{a^8}$$

### B. Power, Zero & Negative Exponents

#### Problem 1

Simplify:  $(x^3)^4$

$$\underline{x^{12}}$$

#### Problem 2

Simplify:  $(2x^2)^3$

$$\underline{8x^6}$$

#### Problem 3

Simplify:  $7^0$

$$\underline{1}$$

#### Problem 4

Simplify:  $(5x^4)^0$

$$\underline{1}$$

#### Problem 5

Simplify:  $3^{-2}$  (write as a fraction)

$$\underline{1/9}$$

#### Problem 6

Simplify:  $x^{-4}$  (write as a fraction)

$$\underline{1/x^4}$$

**Problem 7**

Evaluate:  $-4^2$  vs  $(-4)^2$  — what is  $(-4)^2$ ?

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